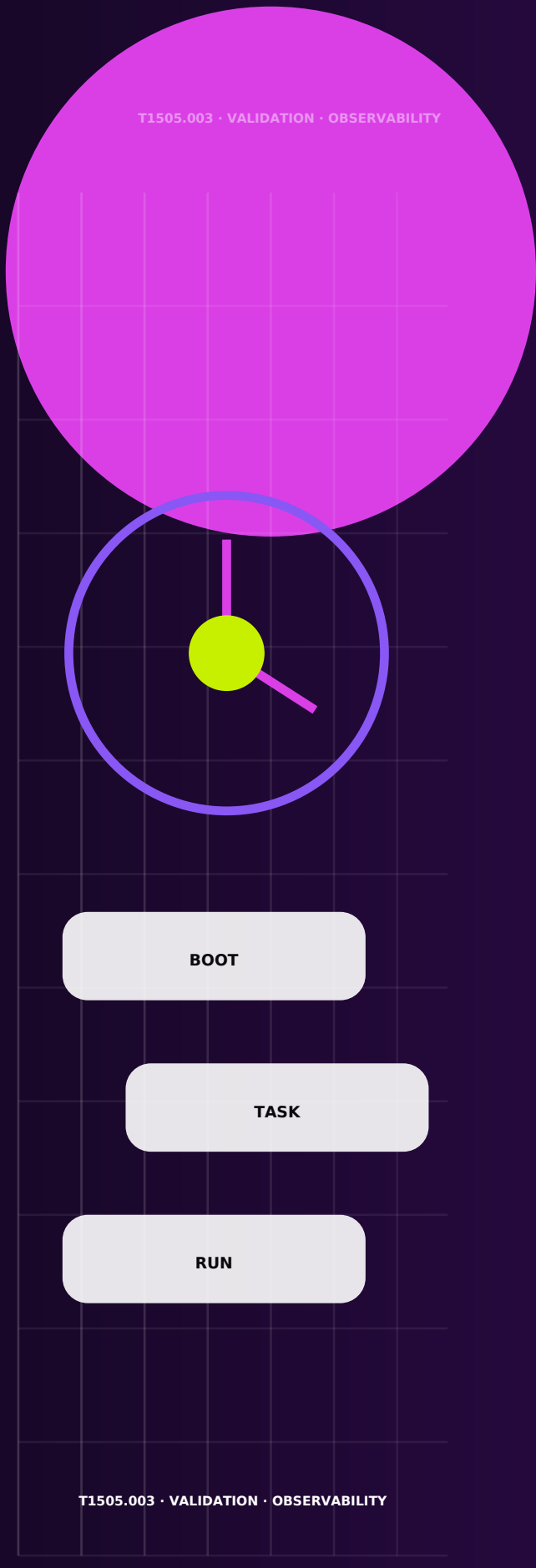
DETECTION THREAT LANDSCAPE

# HERMES HADES - WEBSERVER TO HADOOP SERVICE

Validating source-observed connections from compromised web infrastructure to HiveServer2 and WebHDFS.




VALIDATION QUERY

CONFIDENCE: MEDIUM

REPORT TYPE
campaign

STATUS
Untested implementation sketch

PRIMARY PLATFORM
Microsoft Defender XDR

OPERATIONAL VALUE
8 / 10

01

# Executive Assessment

Decision-grade summary with explicit evidence boundaries.

ASSESSMENT

**DECISION**

Retain this package as a validation query and migrate it to the canonical visual and evidence contract without changing query conditions. Treat every result as an investigative lead until local schema, field population, benign overlap and the legacy claim provenance are reviewed.

### LEGACY EVIDENCE RECORD

- The retained legacy source ledger contains 4 normalized references; claim-to-source mapping remains...

### ANALYST ASSESSMENT

- The existing logic is retained at validation maturity; repository migration is not execution eviden...
- The folder and query header map the package to T1505.003 / T1059; operational relevance still depend...

### DETECTION ASSUMPTIONS

- The named telemetry is collected across the intended asset or identity scope with adequate...
- Projected fields preserve their expected semantics in the target connector and parser. [C002]
- Approved activity, owners and change records are available to analysts during disposition. [C003]

## KEY FINDINGS

### FINDING 1

The original query conditions and source ledger were preserved during layout migration; legacy PDF SHA-256 686698facaf1c9cc... is recorded for traceability. [C001]

### FINDING 2

The principal observable is Evaluate DeviceNetworkEvents for webserver to hadoop service using the existing source-derived conditions while preserving projected investigation context. [C002]

### FINDING 3

The evidence boundary remains explicit: a match does not prove a web shell, successful Hive authentication, UDF creation, command execution, data access, or Hermes/Hades attribution. [C003]

**EVIDENCE BOUNDARY**

A matching result does not prove a web shell, successful Hive authentication, UDF creation, command execution, data access, or Hermes/Hades attribution. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

02

# Threat Mechanics

Technical mechanism, observable transitions and supported scope.

ATTACK FLOW

### TECHNICAL BACKGROUND

The retained dossier links Hermes Hades to the detection opportunity described as webserver to hadoop service. The v2 migration preserves the query exactly and exposes its match boundary, but does not reinterpret legacy source claims as laboratory evidence. Evaluate DeviceNetworkEvents for webserver to hadoop service using the existing source-derived conditions while preserving projected investigation context. [C001][C002]

1

### THREAT CONTEXT

Validating source-observed connections from compromised web infrastructure to HiveServer2 and WebHDFS. Observable: Context retained from the legacy dossier and source ledger.

2

### OBSERVABLE CANDIDATE

Evaluate DeviceNetworkEvents for webserver to hadoop service using the existing source-derived conditions while preserving projected investigation context. Observable: Events from DeviceNetworkEvents and projected fields.

3

### CONTEXTUAL ENRICHMENT

Resolve asset, identity, process, ownership and change context before disposition. Observable: Raw event, surrounding timeline, inventory and authorized-change evidence.

4

### DECISION GATE

Escalate only when corroborating evidence closes the documented non-claim boundary. Observable: Controlled validation, explainable negatives and analyst-readable evidence.

03

# Telemetry & Evidence Model

Minimum data contract and known visibility gaps.

DATA CONTRACT

**PRIMARY PLATFORM**

Microsoft Defender XDR. Evaluate DeviceNetworkEvents for webserver to hadoop service using the existing source-derived conditions while preserving projected investigation context.

| FIELD / TABLE | STATUS | PURPOSE |
| --- | --- | --- |
| **DeviceNetworkEvents** | **Required** | Primary event source selected by the unchanged query logic. |
| **Event time and entity identifiers** | **Required** | Chronology, deduplication, scoping and investigation pivots. |
| **Raw projected context** | **Required** | Preserve the fields emitted by the query before normalization or tuning. |
| **Asset, identity and change inventory** | **Enrichment** | Separate authorized activity and define operational ownership. |

**KNOWN VISIBILITY GAPS**

Unvalidated local schema Missing projected fields Incomplete retention or asset scope Mutable attacker behavior Legacy claim provenance review pending

04

# Primary Detection Logic

Native query, explicit condition evidence and match meaning.

KQL

```
// Rule: [T1505.003][T1059] Hermes Hades - Webserver to Hadoop Service
// MITRE ATT&CK: T1505.003, T1059
// Classification: Validation query
// Status: Untested implementation sketch
// Primary platform: Microsoft Defender XDR
// Required telemetry: DeviceNetworkEvents
// Purpose: Validating source-observed connections from compromised web infrastructure to HiveServer2 and WebHDFS.
// Logic: Evaluate DeviceNetworkEvents for webserver to hadoop service using the existing source-derived conditions while preserving projected investigation context.
// A match proves: An inventoried web server initiated tra□c to an inventoried Hadoop node on a source- observed service port. What it does not prove Web shell, succ
// A match does not prove: a web shell, successful Hive authentication, UDF creation, command execution, data access, or Hermes/Hades attribution.
// Expected false positives: authorized administration, approved software, testing, or benign artifact collisions.
// Coverage limits: local schema, field population, retention, collection scope, and attacker variation require validation.
// Evidence claims: C001, C002, C003
//
// Web server to Hadoop service connectivity validation
// Classification: Validation query - Untested implementation sketch
// Confidence: Medium
// Platform: Microsoft Defender XDR Advanced Hunting
// Required telemetry: DeviceNetworkEvents from every web server in scope
//
// Hunt.io directly documented tooling that used a compromised web server as
// the launch point for HiveServer2 and WebHDFS access on TCP 10000 and 50070.
// Replace both inventories before execution. They are required controls, not
// optional allowlists.
let WebServerDevices = dynamic(["<WEB_SERVER_FQDN>"]);
let HadoopNodeIPs = dynamic(["<HADOOP_NODE_IP>"]);
DeviceNetworkEvents
| where DeviceName in~ (WebServerDevices)
| where RemoteIP in (HadoopNodeIPs)
| where RemotePort in (10000, 50070)
| project
    Timestamp,
    DeviceId,
    DeviceName,
    LocalIP,
    RemoteIP,
    RemotePort,
    RemoteUrl,
    Protocol,
    ActionType,
    InitiatingProcessFileName,
    InitiatingProcessCommandLine,
    InitiatingProcessAccountName,
    InitiatingProcessSHA1,
    ReportId
| order by Timestamp desc

// Interpretation:
// A result proves only that an inventoried web server initiated network
// activity toward an inventoried Hadoop node on a source-observed service port.
// It does not prove a web shell, successful Hive authentication, UDF creation,
// ... complete 58-line query in validation.kql
```

### WHAT A MATCH PROVES

An inventoried web server initiated traffic to an inventoried Hadoop node on a source- observed service port. What it does not prove Web shell, successful authentication, UDF execution, data access, exfiltration or attribution.

### WHAT IT DOES NOT PROVE

a web shell, successful Hive authentication, UDF creation, command execution, data access, or Hermes/Hades attribution.

### MATURITY

Validation query - Untested implementation sketch. Evaluate DeviceNetworkEvents for webserver to hadoop service using the existing source-derived conditions while preserving projected investigation context.

05

# Triage & Hunting Playbook

Evidence-preserving workflow from match to disposition.

WORKFLOW

**01**

Preserve the complete raw matching event, query version, timestamp, connector or sensor version and stable entity identifiers.

**02**

Confirm the event originated from the intended Microsoft Defender XDR data source and verify the semantics of every condition used by the query.

**03**

Resolve asset, identity, process, software and destination ownership against authoritative inventories and approved change records.

**04**

Build the surrounding endpoint, identity, network or appliance timeline and seek an independent evidence layer before escalation.

**05**

Classify benign overlap explicitly; tune only with durable ownership or provenance controls, never by suppressing the core behavior globally.

**06**

Escalate only when corroborating evidence establishes malicious or unauthorized activity beyond the documented match boundary.

**DISPOSITION RULE**

Escalate only when the evidence establishes the behavior described by the query. a web shell, successful Hive authentication, UDF creation, command execution, data access, or Hermes/Hades attribution.

06

# False Positives & Validation

Overlap model, controlled checks and promotion gate.

QUALITY GATE

### AUTHORIZED ADMINISTRATION OR SOFTWARE

Require owner, purpose, approved change and expected execution or configuration provenance.

### BENIGN ARTIFACT OR COMMAND COLLISION

Require the full condition set plus an independent evidence layer or stable provenance.

### PARSER, INVENTORY OR RETENTION AMBIGUITY

Reproduce a controlled event and reconcile raw fields, collection scope and authoritative inventory.

## VALIDATION SEQUENCE

- Generate an authorized controlled equivalent of the primary observable without using live malicious infrastructure.
- Verify each query condition and projected field against retained raw telemetry.
- Run representative approved and near-match negative controls.
- Measure historical prevalence and classify every returned owner or workflow.
- Record parser, connector, retention and asset-scope dependencies with the test evidence.

### SUCCESS

The controlled positive produces a complete, explainable row and representative negatives are separable through durable context.

### REJECTION

Reject promotion when field semantics, collection scope, benign ownership or the core behavior cannot be established reproducibly.

### PROMOTION GATE

Require claim-level source review, a retained controlled positive, representative negatives, measured prevalence and an accountable operational owner.

07

# Analyst Assessment & References

Coverage statement, open questions and source record.

FINAL

**CONCLUSION**

This package now uses the canonical v2 layout and evidence contract while preserving the original validation.kql conditions. It remains a validation query at medium confidence. The visual migration is complete; claim-level provenance and target-environment execution remain explicit promotion gates.

**MITRE ATT&CK**

T1505.003 / T1059

## PRIMARY AND AUTHORITATIVE REFERENCES

**S001 hunt.io - Hermes AI Agent Running Unattended, Hades Implant Staged (2026-07-23)**

hxxps://hunt[.]io/blog/thailand-ministry-finance-targeted-with-hermes-ai-agent

**S002 Microsoft Learn - Microsoft Defender XDR, DeviceNetworkEvents table (undated)**

hxxps://learn[.]microsoft[.]com/en-us/defender-xdr/advanced-hunting-devicenetworkevents-table

**S003 MITRE ATT&CK - T1505.003 - Server Software Component: Web Shell (undated)**

hxxps://attack[.]mitre[.]org/techniques/T1505/003/

**S004 MITRE ATT&CK - T1505.003 - Server Software Component: Web Shell (undated)**

hxxps://attack[.]mitre[.]org/techniques/T1059/

**OPEN QUESTIONS**

A matching result does not prove a web shell, successful Hive authentication, UDF creation, command execution, data access, or Hermes/Hades attribution. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

**DOCUMENT CONTROL**

Schema 2.0 | Layout 2.0 | Maturity validation | Confidence Medium | No customer data | Evidence ledger required